Project Executable Files Manual

User Manual & Installation Guide - Rising Waters

1. Local Development Setup

Installation Steps

1. Extract/Download the project folder to your local system (e.g. `C:\Users\palla\OneDrive\Desktop\rising waters`).

2. Open VS Code ? File ? Open Folder... ? Select the project folder.

3. Open a new terminal inside VS Code.

4. Activate the virtual environment:

```
& ".venv/Scripts/Activate.ps1"
```

5. Install dependencies:

```
pip install -r requirements.txt
```

6. Start the local server:

```
python app.py
```

7. Open your browser and navigate to http://127.0.0.1:5000.

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2. Using the User Interface

Home Screen Dashboard

- View the project introduction, technology stack parameters, and the Machine Learning Classifier comparison table.
- Click Launch Predictor to access the inputs form.

Prediction Inputs Form

- Enter the weather parameters:
- Cloud Cover (0 to 100%)
- Annual Rainfall (in mm)
- Jan-Feb Rainfall (in mm)
- Mar-May Rainfall (in mm)
- Jun-Sep Rainfall (in mm)
- If your inputs violate validation boundaries (e.g. negative precipitation or seasonal totals exceeding annual totals), Bootstrap warning banners will appear. Correct the fields and click Analyze Flood Risk.
- A dark loading spinner block will overlay the form while predictions are executed.

Understanding Outputs

- Green Card (Low Flood Risk): Displayed if parameters indicate normal/safe weather conditions. Displays confidence score progress bar and routine monitoring instructions.
- Red Card (High Flood Risk): Displayed if parameters indicate anomalous rainfall patterns. Displays prediction confidence percentage, input data logs, and recommended evacuation action checklists.